Andy Do

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Education

Cornell University Ithaca, NY

Bachelor of Arts in Computer Science *May 2028*

Relevant Coursework: Machine Learning, Computer System Organization, Discrete Mathematics, Data structures and Algorithms, Linear Algebra, Databases, Functional Programming, Intro to Python

Experience

Fiserv Jun 2026 – Aug 2026

Software Engineer Intern *King of Prussia, PA*

- Migrated a legacy **Java** app with **30+** REST API integrations from VMs to **Kubernetes**, containerizing services with **Docker** and authoring **11+ Helm** resources.
- Automated **GitOps** deployments across environments with **GitLab/ArgoCD**, configuring autoscaling.
- Developed and deployed a **Spring Boot REST API** with **8 endpoints** and **Oracle/JPA** to production, automating merchant onboarding alerts for **20,000+** sales partners.
- Resolved **RBAC** vulnerabilities across **8** Spring controllers using **OAuth2/JWT**, securing payment and signup data for a **50,000+** business network and **500+** users.

Cornell Data Science November 2025 – Present

Machine Learning Engineer *Ithaca, NY*

- Lead a **5**-person ML project team developing a **Mamba-3/PyTorch RAG** assistant to index **138** CDS repositories and past deployed projects to club applicants.
- Configure **CUDA 12.6/PyTorch** workflows for an in-home **NVIDIA GPU cluster**, integrating **Triton/TileLang/CuTe** kernels for Mamba-3 training and inference.
- Built image texture descriptors in **Python** with **OpenCV** and **scikit-image**, benchmarking **161** AI-edited images to identify regions where digital watermarks survive manipulation and presenting findings to **100+** members.

Lancaster Lebanon Intermediate Unit 13 Jun 2024 – Aug 2024

Software Engineer Intern *Lancaster, PA*

- Developed and deployed **2 full-stack applications** using **C#/ASP.NET Core, JavaScript, and SQL Server**, digitizing early-intervention referrals for **3,800+ preschoolers**.
- Managed data for **7000+** parents and caregivers by designing a **SQL** database and implementing backend **CRUD**/business logic for student, guardian, and referral records.

Projects

CodeLens | *Fiserv Hack Week Winner* June 2026

- Built a full stack **Chrome extension** that maps any live web app component to its backend code flow, Jira tickets, and Confluence docs on hover, helping **8** new interns learn their assigned codebases.
- Built a **RAG pipeline** using **ChromaDB** and **Hugging Face embeddings**, integrating **GitHub/GitLab** and Atlassian APIs to trace code execution and retrieve Confluence documentation and Jira tickets.

Lambda | *Next.js, TypeScript, Docker, AWS ECS/Fargate, PostgreSQL* Oct 2025 – Present

- Engineered LLM context compaction with TypeScript, automating token-aware summarization and context replay to support conversations beyond model context limits.
- Overhauled **104 files** across chat infrastructure, implementing streaming, cancellation, image attachments, and persistent conversations with Next.js/Prisma.
- Collaborate on containerizing **2 independently deployable services** using Docker and AWS ECS/Fargate, separating frontend and backend infrastructure.

Technical Skills

Languages: Java, Python, C#, C, JavaScript, TypeScript, SQL, HTML, CSS, OCaml
Frameworks: React, Node.js, NextJS, TanStack, JPA, Spring Boot, tRPC, Bootstrap, Flask, ASP.NET Core, SQLAlchemy, Prisma
Libraries: Pandas, NumPy, Matplotlib, scikit-learn